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Estimating the pressure-limited CO_2 injection and storage capacity of the United States saline formations: Effect of the presence of hydrocarbon reservoirs

Hossein Jahediesfanjani[a,*], Peter D. Warwick[b], Steven T. Anderson[b]

[a] Lynxnet, LLC, Suffolk, VA, United States
[b] U.S. Geological Survey, Reston, VA, United States

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ABSTRACT

The U.S. Geological Survey (USGS) national assessment of carbon dioxide (CO_2) storage capacity evaluated 192 saline Storage Assessment Units (SAUs) in 33 U.S. onshore sedimentary basins that may be utilized for CO_2 storage (see USGS Circular 1386). Similar to many other available models, volumetric analysis was utilized to estimate the initial CO_2 injection and storage capacity of these SAUs based on aquifer characteristics and buoyant and residual trapping. The factor being almost always overlooked in most CO_2 storage capacity models is that many of the evaluated SAUs contain large numbers of both conventional and unconventional discovered and undiscovered oil and gas reservoirs. The hydrocarbon production and pressure distribution of the resident oil and gas reservoirs may be negatively influenced by the propagated CO_2 plume and pressure front resulting from a CO_2 injection and storage operation in the surrounding SAU.

To have a more realistic and accurate estimation of CO_2 injection and storage capacity in saline formations, a model was previously developed that considers the CO_2 injectivity of a given formation, underground pressure build-up limitations imposed by the rock fracturing pressure and the presence of hydrocarbon reservoirs within these aquifers. The developed method estimates the pre–brine extraction, pressure-limited CO_2 injection and storage capacity of a saline formation by applying 3D numerical simulation only on the effective injection area (A_{eff}) surrounding each CO_2 injection well utilizing TOUGH2-ECO2N simulation software.

In this work, we have identified and accounted for the existence of all hydrocarbon-bearing formations within a selected SAU by designating a buffer zone of no CO_2 injection and no induced pressure buildup around each hydrocarbon reservoir and only injecting CO_2 in the remaining SAU volume. Applying previously developed method, the pressure-limited CO_2 storage capacities of all U.S. SAUs are estimated in three scenarios: 1) ignoring the presence of oil and gas reservoirs, 2) accounting for the existence of oil and gas reservoirs, and 3) injecting CO_2 only in the SAUs without significant numbers of oil and gas reservoirs (defined as less than half of their area being covered by hydrocarbon-bearing formations). Under each of these three scenarios, the number of required wells, well spacing, and corresponding CO_2 injection rates are estimated and presented in detail for every SAU defined by the USGS national CO_2 storage assessment. Using this explained approach, the results indicate that the non-hydrocarbon sections of hydrocarbon-bearing SAUs in conjunction with other non-hydrocarbon-bearing SAUs may be capable of injecting and storing about 1.0 billion metric tonnes (gigatonnes, Gt) of CO_2 per year in the United States for 50 years without requiring pressure management techniques, such as brine extraction. The current approach can also be generally applied to any other saline formation with characteristics similar to those of the defined SAUs.

1. Introduction

Carbon dioxide (CO_2) injection and storage in underground saline formations is one of the most important methods to reduce CO_2 emission into the atmosphere. Even though most climate change prevention scenarios recommend developing a plan to replace fossil fuels with renewable, non-polluting energy sources worldwide (Energy Technology Perspective (ETP, 2012), it is predicted that fossil fuels will still play a major role in the energy sector for decades, mainly because it will not be possible to switch to completely non–fossil fuel energy production in a timely manner and at a reasonably acceptable cost

* Corresponding author.
E-mail address: hjahediesfanjani@usgs.gov (H. Jahediesfanjani).

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(European Commission, 2013). Continued use of fossil fuels makes the problem of produced greenhouse gases (GHGs) and especially CO_2 emission into atmosphere and associated harmful effects on the climate very important.

In this regard, International Energy Agency (IEA) proposed two climate-change prevention scenarios: 1. 2 ° Celsius (2DS) and 2. Beyond 2DS (B2DS) that tend to suggest practical ways to reduce emission of GHGs into atmosphere to limit average global temperature change to 2 °C for 2DS and below 2 °C for B2DS scenarios (IEA, 2017). Based on the reported evaluations, the CO_2 Capture and Storage (CCS) will be responsible for up to 14% of reduction of global CO_2 emissions to meet 2DS and an additional 32% of the path from 2DS to B2DS by 2060 and beyond (IEA, 2017). Practically, it means that worldwide, by 2060, on average approximately 7.0 billion metric tonnes (gigatonnes, Gt) of CO_2 must be captured and stored each year to meet B2DS (IEA, 2017), as also emphasized at the 2015 Paris Climate Conference (Paris climate report, 2016). According to IEA forecasts, the United States should store 1.0 Gt/year of CO_2 (IEA, 2017). Note that the mentioned estimate is only based on the B2DS requirements. The current annual GHGs emissions in the United States are approximately 7.0 Gt of CO_2 equivalents (EPA, 2015), of which a high percentage would have to be eliminated by energy fuel switching and transformations to meet the B2DS requirements.

In this regard, CO_2 from anthropogenic sources used for enhanced oil recovery (EOR) has gained some attention during the last few years because the CO_2-EOR process produces additional oil from oil reservoirs while allowing some CO_2 to be stored within the reservoir structure. However, the additional oil recovery due to the CO_2-EOR operation is usually in the range of 5 to 15% of the original oil in place (OOIP) (Jahediesfanjani, 2017) for each amenable reservoir, so the corresponding CO_2 storage capacity volume of these reservoirs is at least an order of magnitude less than the billions of metric tonnes required for effective climate change mitigation (USGS CO_2 assessment team, 2013a, 2013b, and NETL, 2012).

Underground saline formations have been recommended as an alternative site for CO_2 storage due to their widespread availability and considerable capacity to hold and store CO_2.

Assuming that the required technology and infrastructure exists to capture and transport CO_2, an important concern is to estimate the most likely volume of CO_2 that can be injected and stored annually in U.S. underground saline formations, and to determine whether that volume is sufficient to meet the B2DS requirements (IEA, 2017). To more realistically and accurately model and estimate CO_2 injection and storage capacity in saline formations, we have developed a model that takes into account some limitations associated with the CO_2 injection into a given formation such as underground pressure build-up limitations imposed by the rock fracturing pressure and the presence of hydrocarbon reservoirs within these aquifers. The developed method estimates the pressure-limited CO_2 injection and storage capacity of a saline formation by applying 3D numerical simulation only on the effective injection area (A_{eff}) surrounding each CO_2 injection well utilizing TOUGH2-ECO2N simulation software. The result of applying this method is considered as further refinement of the estimates by USGS of CO_2 storage capacity in saline formations in the United States taking into account underground pressure limitations.

The next step in the process is to plan the number, spacing, and injection rates of CO_2 injection wells on a national level to safely accommodate the needed CO_2 underground storage capacity. The estimated storage capacity needs to be matched with large stationary sources that will be subject of one of our future investigations.

2. Literature review

Many researchers have developed models to estimate the ultimate capacity of a given saline formation to store and hold CO_2 (Bandilla et al., 2015). These models use different assumptions and understandings of the entire phenomenon, which results in highly variable estimates of potential storage capacity (Heidug, 2013). Many of the models estimate the CO_2 storage capacity of a given formation based on its pore volume and a CO_2 storage efficiency value (volumetric models). Bachu (2015) reviewed many of these models and evaluated the effects of various parameters on the estimated CO_2 storage efficiency under different reservoir and boundary conditions. These models define the saline formation as a homogeneous porous medium with uniform properties in which hydrodynamic and buoyancy forces must overcome viscous and capillary forces for the injected CO_2 plume to propagate into and through the saline formation. It is ultimately the rock, brine, and CO_2 conditions and properties that determine which force dominates and may affect the ultimate CO_2 storage efficiency. The estimated CO_2 storage efficiency is eventually used with the reservoir pore volume to estimate the CO_2 storage capacity of a saline formation (Celia et al., 2015).

There are several models in the literature that each report a different methodology to estimate saline formations' CO_2 injection and storage capacity at national or semi-national level. Some of these models are discussed here. For example, the U.S. Department of Energy (DOE) and the U.S. Geological Survey (USGS) have each developed a distinct, comprehensive method to estimate the national CO_2 storage capacity in saline formations by developing a different method to estimate ranges of appropriate CO_2 storage efficiency values. These models, along with the Szulczewski et al. (2012) model and a few others such as the Eccles et al. (2012) model, are reviewed below. All of these models are developed based on somewhat different assumptions and considerations and are fundamentally different from our proposed pressure-limited dynamic model that is based on CO_2 injectivity and underground fracture pressure limitations (Jahediesfanjani et al., 2017). Nevertheless, because these reports have reported CO_2 injection and storage capacity of major U.S. onshore saline formations, they are reviewed below.

As part of a national study, the U.S. DOE developed a method to estimate the capacity of the saline formations to hold and store CO_2 (Goodman et al., 2011) They incorporated the volumetric method to estimate the ultimate CO_2 capacity of a given saline formation, and estimated CO_2 storage efficiency assuming that net-to-gross thickness, effective-to-total porosity, volumetric displacement, and microscopic displacement can be represented by log-odds normal distributions at regional and national levels. Monte Carlo simulations were used to estimate saline formation storage efficiencies ranging from 0.40 to 5.5% for three different lithologies over the 10% and 90% probability ranges, respectively. The CO_2 capacity of a saline formation can be estimated by multiplying its estimated storage efficiency by its estimated pore volume. Utilizing the developed approach, the CO_2 storage capacity of saline formations throughout the United States was estimated to be 2300 and 22,000 Gt for the low and high estimates, respectively (NETL, 2012).

A similar method was developed by the USGS (USGS CO_2 assessment team, 2013a and 2013b) to estimate the CO_2 storage capacity of 192 storage assessment units (SAUs) in 33 sedimentary basins throughout the United States. In this previous model, the USGS national assessment team accounted for two mechanisms for CO_2 storage after injection: buoyant and residual trapping, each with a very different range of storage efficiency values. The model estimated fixed buoyant storage efficiencies using petroleum field observations and injection histories. The residual storage efficiencies were estimated using a relationship based on a capillary trapping number and mobility factor calculations, which required knowledge of CO_2 and brine residual saturations, viscosities, and relative permeability values (Brennan, 2014). Monte Carlo simulations (Blondes et al., 2013) were used to estimate the effect of uncertain parameters. The results were published as ranges of CO_2 storage capacity for each saline formation, with total national CO_2 storage capacity for all 192 SAUs ranging from 2000 to 4000 Gt based on 5% and 95% probability (P_5 and P_{95}) analyses, respectively

(USGS CO_2 assessment team, 2013a and 2013b).

Szulczewski et al. (2012) developed a model based on residual and solubility trapping of CO_2 during and after injection into a saline formation along a narrow fairway within a geological basin. They derived a one-dimensional partial differential equation assuming a homogeneous saline formation and flow that is in vertical equilibrium at any given time and location of the CO_2 plume. They estimated two types of CO_2 capacity—pressure-limited and migration (space)-limited—for 11 saline formations. The pressure-limited capacity is a dynamic value of the total amount of CO_2 that can be injected into the formation without causing formation fracture or activation of existing faults. The estimated volume grows over time, and at its maximum equals the migration-limited capacity volume. They estimated a total volume of approximately 314 Gt of CO_2 that can be injected and stored for up to 600 years in the 11 studied formations. The Mount Simon Formation had the largest storage capacity, estimated at 111 Gt of CO_2, followed by the Black Warrior River Aquifer, estimated at 90 Gt.

Others have also applied mainly analytical and semi-analytical models to provide CO_2 injection and storage capacities of some saline formations. For example, Eccles et al. (2012) estimated the CO_2 storage capacity of 13 saline formations in the U.S. to be over 925 Gt by applying an analytical expression. The formation with the greatest estimated storage capacity was the Frio Formation of the Gulf Coast Basin (322 Gt), followed by the Mount Simon Sandstone of the Illinois Basin (254 Gt). These numbers show highly variable ranges of estimated CO_2 capacity of these formations when compared to 3D numerical models that estimate the dynamic CO_2 capacity of a formation in the presence of limitations on pressure buildup in the reservoir and considering formation heterogeneities. For example, the estimated pressure-limited CO_2 storage capacity of the Mount Simon Sandstone is 5 Gt using the Zhou et al. (2010) model, 3.5 Gt using the Jahediesfanjani et al. (2017) model, 111 Gt using the Szulczewski et al. (2012) model, 254 Gt using the Eccles et al. (2012) model, 11–150 Gt (42.2 Gt based on the medium scenario) using the NETL (2012) model, and 62–150 Gt (91 Gt based on the 50% probability, or P_{50}, scenario) using the USGS model (USGS CO_2 assessment team, 2013a and 2013b).

Comparing these methods and produced CO_2 storage capacity values indicates that the total CO_2 injection capacity of a saline formation is influenced by at least two factors. 1. Different aquifer size, and geological and petrophysical properties, such as porosity, permeability, depth and thickness values, are used in different methods for characterizing the same aquifer. 2. Some of these methods are built upon considering the effect of underground dynamic pressure build-up and its associated limitations in the injected CO_2 rate and volume, while many of the other models are developed based on volumetric assumptions. The latest group of models is based on estimating formation storage efficiency values that use only aquifer characteristics and do not consider dynamic pressure build-up limitations.

The models that do not consider underground aquifer pressure build-up and fracture pressure limitations provide theoretical and technically accessible capacities that can only be achieved under certain conditions (CSLF, 2008). Some of these conditions are: open reservoir boundaries; homogeneous and isotropic reservoir properties; no limit on time of continuous CO_2 injection; no limit on the number of CO_2 injection wells or their spacing; no faults, natural barriers, or other discrepancies so that pressure does not build up; and no economic or technical limitations. Practically, many factors affect the ultimate CO_2 storage capacity of a given saline formation. One of the important factors is the dynamic reservoir pressure (Eccles et al. (2012); Zhou et al., 2010; Bandilla et al., 2012; and Senel and Chugunov, 2013). During CO_2 injection, the reservoir pressure front induced around an injection well begins propagating from the wellbore toward the boundaries of the reservoir. An important consideration is that the maximum reservoir pressure needs to stay below a certain percentage of the formation fracture pressure to avoid fracturing the formation rock. Therefore, the well spacing, injection rate, and injection duration must be optimized to meet the mentioned important criterion.

The other important consideration is what reservoir boundary conditions to use. Theoretically, if basin-wide, industrial-scale CO_2 injection were implemented, the entire saline formation would need to be developed by a series of CO_2 injection wells with a certain (usually equal) well spacing. Therefore, a saline formation may be populated by hundreds of wells depending on the formation area and prevailing reservoir conditions (Jahediesfanjani et al., 2017; Mathias et al., 2011; Yang et al., 2015a; and 2015b, and Azizi and Cinar, 2013). In our model, we first populate the entire saline formation with equally spaced CO_2 injection wells (25 km by 25 km spacing) and assign a CO_2 injection rate to each injector. The assigned injection rate is estimated by 3D numerical modeling of CO_2 injection throughout the saline formation based on the specific geological characteristics (porosity, permeability, depth, and thickness) of the selected section of the saline formation, with the constraint that the pressure can never exceed 80% of the reservoir fracture pressure (Jahediesfanjani et al., 2017). The model utilizes TOUGH2-ECO2N reservoir simulation software (Pruess, 2005). The CO_2 injection rates are designed so that the induced pressure front from each well will practically be able to propagate only until it reaches a virtual no-flow boundary, half way between injection wells, after 50 years of injection (the pressure increase at cell boundary at the end of 50 years of CO_2 injection will be less than 1 bar). It is important to note that, this is an artificially imposed boundary condition so that neither pressure nor CO_2 plume does not propagate from one cell to another. If the saline formation is assumed to be homogeneous and of constant thickness, the CO_2 injection rate for all wells will be practically equal. More details regarding the model that is developed and utilized as an alternative to 3D full-field numerical simulation, are explained in full elsewhere (Jahediesfanjani et al., 2017).

3. Using the model to predict CO_2 storage capacity of saline formations

For each investigated SAU, the USGS national CO_2 storage assessment (USGS CO_2 assessment team, 2013a,b) evaluated and reported the minimum, most likely, and maximum values for 1) reservoir porosity, 2) reservoir permeability, 3) reservoir total thickness, 4) reservoir net thickness, 5) reservoir depth, 6) reservoir area, and 7) the fraction of the effective reservoir area with groundwater salinity over 10,000 mg/L (in the U.S. water with salinity less than 10,000 mg/L is groundwater protected for human consumption). In estimating the ultimate pressure-limited dynamic SAU CO_2 storage capacity, all these parameters are important and must be considered.

The Jahediesfanjani et al. (2017) model was utilized to estimate the 50-year CO_2 injection capacity for each well, called the CO_2 injection density (C_{ID}, metric tonnes of injected CO_2 per year per well) based on cell sizes of 25 km by 25 km and duration of injection of 50 years.

This is a 3D model with the third dimension (Z-dimension) being equal to the thickness of each cell that is estimated from net thickness spatial maps for each cell within each SAU. In other words, consider a cell whose X–Y dimensions are fixed at 25 km by 25 km while the Z-dimension is allowed to vary based on the net thickness of the cell at the given location. We have created various realizations in which X–Y dimensions are fixed for each cell while other properties such as porosity, permeability, depth, and net thickness vary from a minimum to maximum value. The 3D numerical simulation runs are performed for each realization (108 total realizations) utilizing Tough2-ECO2N software to estimate the maximum CO_2 injection capacity for each cell (CO_2 injection density) under the pressure limitation conditions. The results are fed into a Visual Basic (VB) program in order to perform necessary extrapolations and modifications. For each cell within a given SAU, if we have local properties, the VB program estimates the maximum CO_2 injection capacity based on the simulated results, adds up the estimated results for all the cells, and reports the final value for a given SAU as the maximum CO_2 injection capacity. This procedure has been described in

full details in our previous paper with an example (Mount Simon Sandstone in Illinois Basin) (Jahediesfanjani et al., 2017). This procedure is repeated and described in more details for the Sligo, Hosston, and Cotton Valley Formation SAU in Gulf of Mexico in Appendix I.

CO_2 injection density indicates how well or poorly a portion of a formation receives and stores injected CO_2 while remaining within an acceptable pressure range. Certainly, the major goal of any commercial CO_2 injection and storage project would be to optimize C_{ID} based on the prevailing reservoir conditions so that the maximum annual volume of CO_2 per well is injected while the maximum reservoir pressure is kept below 80% of the reservoir fracture pressure.

The value of C_{ID} used by Zhou et al. (2010) is five million metric tonnes (Mt) of CO_2 per year per project; that of Senel and Chugunov (2013) based on the Illinois Basin Decatur Project, is approximately 335,000 metric tonnes of CO_2 per year per well; both studies were performed for the Mount Simon Formation of the Illinois Basin. In our study the minimum, most likely, and maximum optimum C_{ID} is obtained for each SAU using the generalized type curves of Jahediesfanjani et al. (2017). For evaluation and comparison purposes, the most likely estimated values of C_{ID} are then classified into the six groups shown in Table 1. The classification is arbitrary and is based on our experience in evaluation of CO_2 injection capacity under different saline formation geological and petrophysical properties that are used to estimate and report the ultimate most likely annual and 50-year CO_2 storage capacity of each SAU. However, due to economics or development limitations, operators will tend to develop those SAUs with higher values of C_{ID} (Groups 3 and higher in Table 1) over poorly performing SAUs (Groups 1 and 2 in Table 1). The process of estimating the pressure-limited dynamic CO_2 injection and storage capacity under a series of conditions and scenarios is fully described below for the U.S. Gulf of Mexico Basin.

3.1. Pressure-limited (dynamic) CO_2 injection and storage capacity of the U.S. Gulf of Mexico basin

Based on the USGS assessment (USGS CO_2 assessment team, 2013a and 2013b), the U.S. Gulf of Mexico Basin within the U.S. portion of the Gulf Coastal Plain contains 27 SAUs, with a total area of approximately 2.7 million km^2 (667.1 million acres) available for CO_2 storage. These SAUs, summarized in Table 2, cover most of southern and eastern Texas, all of Louisiana, much of Mississippi, southern Arkansas, southern Alabama, and limited parts of Florida and Georgia.

The USGS-assessed (USGS CO_2 assessment team, 2013a and 2013b) most likely values of depth, net thickness, porosity, permeability, and other properties were used to obtain the CO_2 injection parameters for the 27 SAUs in the U.S. Gulf of Mexico Basin. Table 2 summarizes the results of the calculation procedure and shows estimated C_{ID}, the corresponding injection group, total SAU annual injection rate (V_t), number of wells required to inject the estimated volumes (N_w), and the 50-year total CO_2 storage capacity of each SAU. It was not possible to estimate CO_2 injection and storage parameters for the Knowles and Winn Limestones and Calvin Sandstone SAU and the Tertiary Slope and Basin Floor SAU because the USGS national assessment (USGS CO_2 assessment team, 2013a,b) did not report all the properties necessary for the calculation procedure. Based on Table 2 and utilizing the Jahediesfanjani et al. (2017) model, the total annual CO_2 injection for the remaining 25 SAUs in the U.S. Gulf of Mexico Basin is estimated to be around 1432 Mt/year, if all of these formations are fully developed. The full development of these SAUs would require drilling 3766 injection wells in equal 25 km by 25 km spacing. It would allow injection and storage of 71.6 Gt of CO_2 over the injection duration of 50 years.

As shown in Table 2, note that 12 of the 25 SAUs are in Group 1, with an average C_{ID} of only 30,000 metric tonnes of CO_2 per year per well. This means 1570 injection wells would need to be drilled and operated to inject only 42 Mt of CO_2 per year. Because of very low injection capacity these SAUs were eliminated from further consideration. Similarly, an additional 4 of the remaining 13 SAUs are in Group 2 with an average C_{ID} value of only 145,000 metric tonnes of CO_2 per year per well. This indicates that 1090 CO_2 injection wells are required to inject only 140 Mt of CO_2 per year, which also might be considered lower than industrial practice. The C_{ID} values for these 16 SAUs are based on the most likely values for reservoir properties and some areas of these SAUs might have higher C_{ID} values. However, because of the lack of sufficient data to investigate the matter, we base our analysis and screening process on the most likely values. Consequently, we do not recommend these SAUs for CO_2 injection and storage, unless a relatively efficient pressure management technique is planned, such as brine extraction before or during CO_2 injection.

Out of the remaining nine SAUs, three are in Group 3, with an average C_{ID} of 380,000 metric tonnes per year per well; four are in Group 4, with an average C_{ID} of 680,000 metric tonnes per year per well; one is in Group 5, with C_{ID} of 1,820,000 metric tonnes per year per well; and one is in Group 6 with a C_{ID} of 2,750,000 metric tonnes per year per well. These nine SAUs have moderate to high values of C_{ID}, which characterizes them as the best candidates for CO_2 injection and storage. Our investigation estimates that 1110 CO_2 injection wells are required to inject approximately 1.25 Gt of CO_2 per year for 50 years, resulting in approximately 62.5 Gt of CO_2 storage without requiring pressure management techniques such as brine extraction.

It is important to consider that the estimated volumes are obtained based on the most likely property values for each SAU and, in reality, reservoir property distribution and heterogeneity may dictate certain CO_2 injection development strategies throughout each SAU. In other words, there may be some sweet and favorable spots within each SAU that may be able to handle higher than the estimated CO_2 injection rates, and some zones with poorer properties that may disqualify them from development for CO_2 injection. For further investigation and to determine what portion of each SAU is appropriate for CO_2 injection and storage development considerations, the mentioned nine most favorable SAUs were studied in more detail. Due to space constraints, the mentioned process is described here only for the Sligo and Hosston Formations and Cotton Valley Group SAU, which is the most important SAU in the U.S. Gulf of Mexico Basin with the greatest CO_2 injection and storage potential.

Maps of depth and thickness for all studied SAUs were created using data from the USGS national assessment (USGS CO_2 assessment team, 2013a,b) in order to generate CO_2 injection density maps. The developed process is explained in detail for the Sligo and Hosston Formations and Cotton Valley Group SAU in Appendix I.

Figs. 1 and 2 present the CO_2 injection density (C_{ID}) of the Sligo and Hosston Formations and the Cotton Valley Group, respectively, in the Sligo and Hosston Formations and Cotton Valley Group SAU. Part of the area, indicated in pink, shows very low estimated C_{ID}. The rest of the SAU area, with C_{ID} of at least 250,000 metric tonnes of CO_2 per year per

Table 1
Ranges of CO_2 injection density (C_{ID}) values, in Mega tonnes (Mt)/ year/ well, classified into six groups.

	Very poor Group 1	Poor Group 2	Acceptable Group 3	Moderate Group 4	Rich Group 5	Very rich Group 6
Minimum (greater or equal) Mt/year/well	0	0.1	0.25	0.50	1.0	2.5
Maximum (less than) Mt/year/well	0.1	0.25	0.50	1.0	2.5	–

Table 2
Injection rate per year per well (C_{ID}), total storage assessment unit (SAU) injection rate, and number of wells required for each SAU in the U.S. Gulf of Mexico Basin. SAUs and most likely input calculation values are from U.S. Geological Survey Geologic Carbon Dioxide Storage Resources Assessment.[4,5].

SAU Code	SAU Name	C_{ID}, Mt/year /well	Group	V_t, Mt/year	N_w	50-year CO_2 Storage Capacity, Gt
C50490102	Norphlet Formation Deep	0.02	1	6.46	272	0.32
C50490103	Smackover Formation	0.03	1	6.01	176	0.3
C50490104	Smackover Formation Deep	0.03	1	9.5	380	0.47
C50490105	Haynesville Formation	0.02	1	1.58	92	0.08
C50490106	Haynesville Formation Deep	0.02	1	0.85	57	0.04
C50490110	Rodessa Formation and James Limestone	0.02	1	3.51	181	0.18
C50490111	Rodessa Formation and James Limestone Deep	0.01	1	0.61	56	0.03
C50490113	Edwards, Glen Rose, and James Limestones	0.02	1	0.39	23	0.02
C50490114	Washita and Fredericksburg Groups, Rusk Formation, and James Limestone	0.01	1	1.07	157	0.05
C50490115	Washita and Fredericksburg Groups, Rusk Formation, and James Limestone Deep	0.01	1	0.27	27	0.01
C50470119	Queen City Sand	0.07	1	3.37	51	0.17
C50470120	Sparta Sand	0.09	1	8.33	98	0.42
C50490101	Norphlet Formation	0.11	2	20.25	189	1.01
C50490112	Fredericksburg Group and Rusk Formation	0.17	2	6.79	108	0.34
C50490116	Tuscaloosa and Woodbine Formations	0.12	2	62.41	510	3.12
C50490117	Navarro, Taylor, and Austin Groups	0.18	2	50.46	279	2.52
C50490108	Sligo and Hosston Formations and Cotton Valley Group Deep	0.27	3	37.3	129	1.9
C50470118	Carrizo Sand and Wilcox Group	0.48	3	96.64	246	4.83
C50470121	Yegua and Cockfield Formations	0.39	3	47.49	121	2.37
C50470122	Frio and Vicksburg Formations	0.73	4	111.83	154	5.59
C50470123	Lower Miocene I	0.73	4	39.8	55	1.99
C50470124	Lower Miocene II	0.68	4	40.79	62	2.04
C50470125	Middle Miocene	0.58	4	13.67	24	0.68
C50470126	Upper Miocene	1.82	5	22.8	13	1.14
C50490107	Sligo and Hosston Formations and Cotton Valley Group	2.75	6	839.88	306	41.99
C50490109	Knowles and Winn Limestones and Calvin Sandstone	N/A	N/A	N/A	N/A	0
C50470127	Tertiary Slope and Basin Floor	N/A	N/A	N/A	N/A	0
Total, Group 1				41.95	1570	2.10
Total, Group 2				139.91	1086	7.00
Total, Group 3				181.43	496	9.07
Total, Group 4				206.09	295	10.30
Total, Group 5				22.8	13	1.14
Total, Group 6				839.88	306	41.99
Total, Groups 1–6				1432.06	3766	71.60
Total, Groups 3–6				1,250.2	1110	62.51

C_{ID}: CO_2 injection density, Mt/year/well; V_t: total CO_2 injection volume in SAU per year, Mt/year; N_w: number of wells required to inject CO_2 in each SAU; N/A: not applicable.

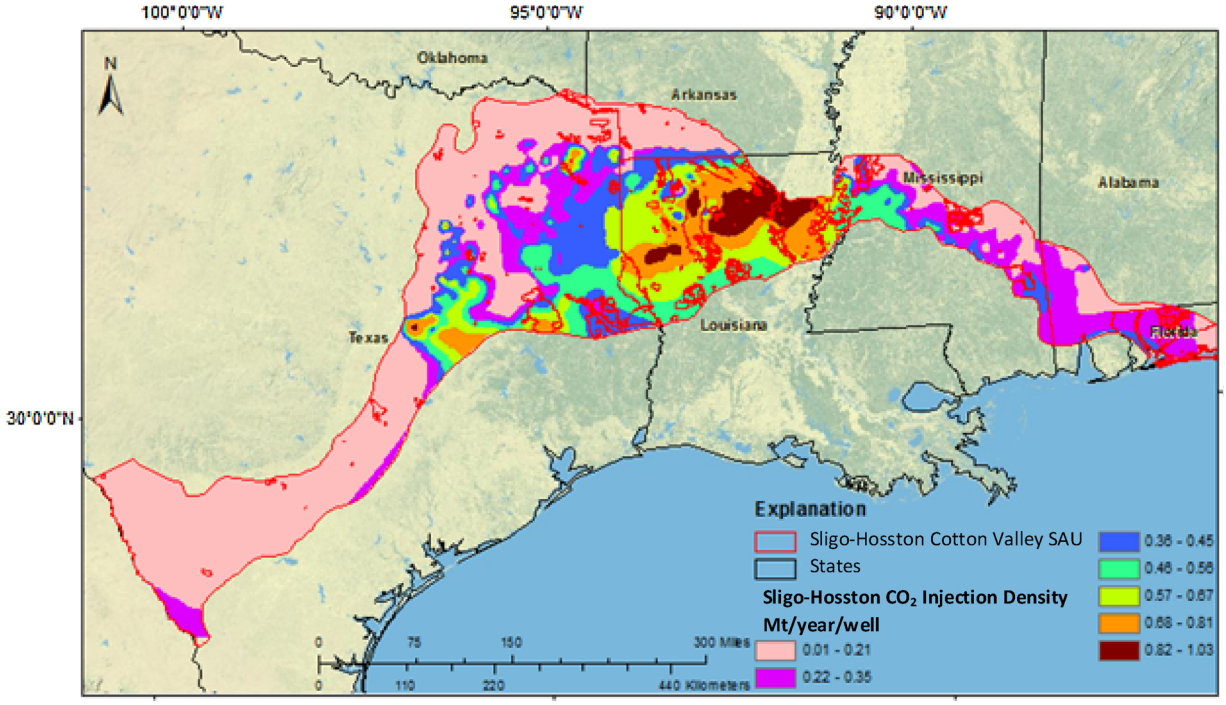


Fig. 1. CO_2 injection density (C_{ID}) map for Sligo and Hosston Formations in the Sligo and Hosston Formations and Cotton Valley Group SAU (storage assessment unit).

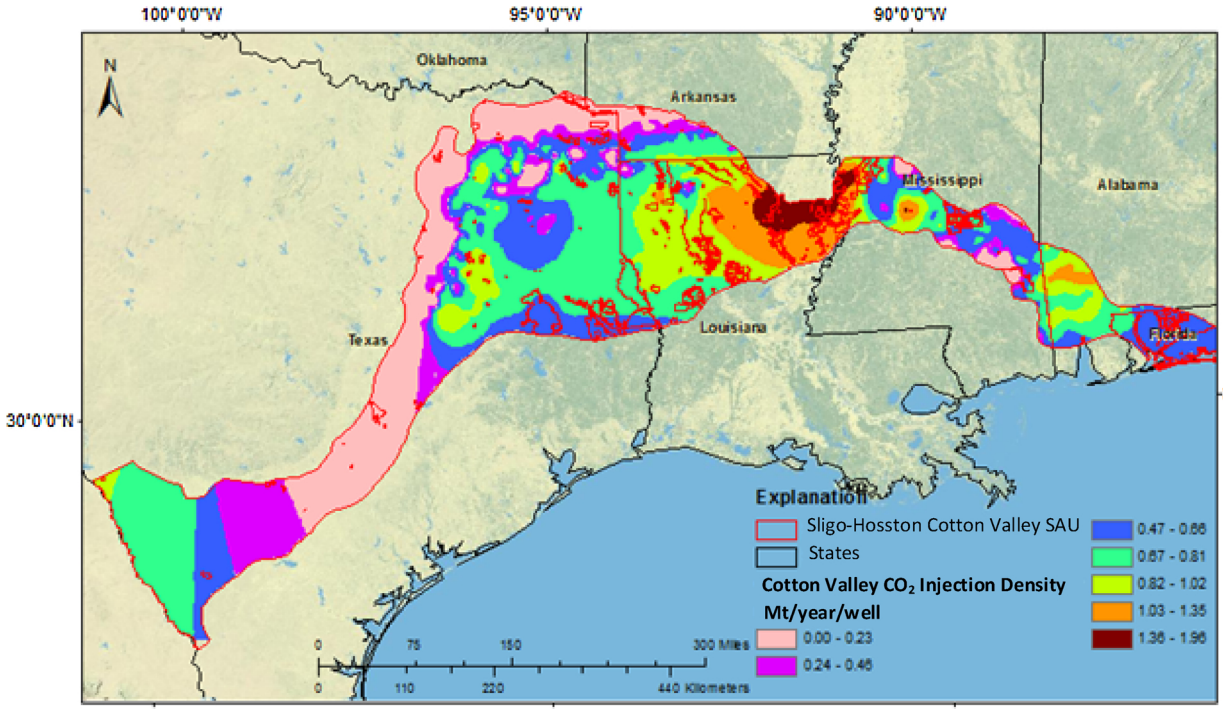


Fig. 2. CO_2 injection density (C_{ID}) map for the Cotton Valley Group in the Sligo and Hosston Formations and Cotton Valley Group SAU (storage assessment unit).

well, can be developed for CO_2 injection and storage.

Figs. 3 and 4 assess the investigated issue further by presenting cumulative area coverage and cumulative CO_2 injection versus CO_2 injection rate for the Sligo and Hosston formations and the Cotton Valley Group, respectively. If the minimum acceptable CO_2 injection rate is chosen to be 250,000 metric tonnes per year per well (Group 3 and higher), based on Fig. 3, approximately 46% of the Sligo and Hosston formations part of the Sligo and Hosston Formations and Cotton Valley Group SAU may be disqualified for CO_2 injection development. This corresponds to approximately 17.5% of cumulative CO_2 injection. It implies that approximately 114,500 km^2 of the SAU is eligible to be considered for CO_2 injection, yielding a CO_2 injection and storage capacity of over 194.0 Mt/year using 183 CO_2 injection wells. Similarly, based on Fig. 4, approximately 18% of the Cotton Valley Group part of the Sligo and Hosston Formations and Cotton Valley Group SAU may be disqualified for CO_2 injection development; the disqualified area corresponds to less than 3.5% of cumulative CO_2 injection.

Based on the above-mentioned procedure approximately 186,500 km^2 of this SAU is eligible to be considered for CO_2 injection, yielding a CO_2 injection and storage capacity of over 551.0 Mt/year using 296 CO_2 injection wells. Overall, both formations together can

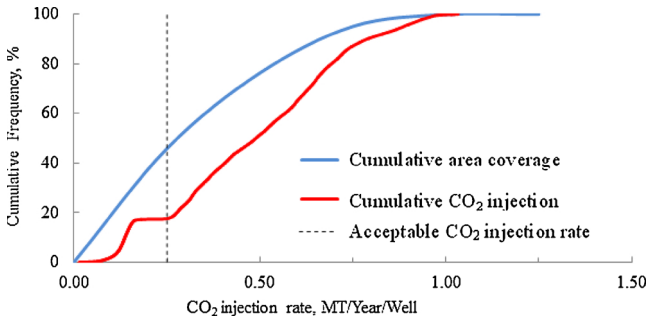


Fig. 3. Cumulative area coverage and cumulative CO_2 injection versus CO_2 injection rate for Sligo and Hosston Formations in the Sligo and Hosston Formations and Cotton Valley Group SAU (storage assessment unit).

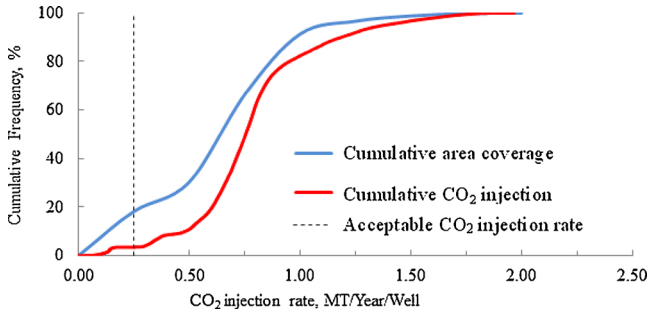


Fig. 4. Cumulative area coverage and cumulative CO_2 injection versus CO_2 injection rate for the Cotton Valley Group in the Sligo and Hosston Formations and Cotton Valley Group SAU (storage assessment unit).

accommodate roughly 745.0 Mt/year of CO_2 injection via 296 injectors that will result in approximately 37.25 Gt of CO_2 storage capacity in 50 years of continuous CO_2 injection compared with 41.99 Gt obtained by applying most likely values of depth, net thickness, porosity and permeability (see Table 2).

This process was repeated for the eight other important SAUs in the U.S. Gulf of Mexico Basin. Table 3 summarizes the acceptable annual CO_2 injection capacity, number of required wells, and total 50-year CO_2 storage capacity for all nine selected SAUs. Based on Table 3, 1100 Mt of CO_2 can be injected and stored in parts of these SAUs that show the storage capacity of over 250,000 metric tonnes of CO_2 per year. Approximately 965 wells are required to inject the required amount of CO_2 and the ultimate 50-year CO_2 storage capacity of these SAUs together is over 55.5 Gt of CO_2. It shows a 13% and 11% drop, respectively, in the number of injectors and 50-year CO_2 storage capacity of these formations compared to the values reported in Table 2.

Our investigation based on oil and gas reservoir production from IHS Inc. (2012); Nehring Associates Inc. (2012), and the Comprehensive Resource Database (CRD) (Carolus et al., 2017) indicates that many of the investigated SAUs in the U.S. Gulf of Mexico Basin contain a large number of both abandoned and active oil and gas reservoirs. For example, Fig. 5 shows the areal extent of the oil and gas reservoirs in the

Table 3
SAU (storage assessment unit) injection capacity evaluation showing acceptable CO_2 injection ranges, required numbers of wells, and 50-year CO_2 storage capacities. SAUs and most likely input calculation values are from U.S. Geological Survey Geologic Carbon Dioxide Storage Resources Assessment Team, 2013a; and 2013b.

SAU Name	A_a, %	I_a, %	V_t, Mt/year	N_w	50-year CO_2 Storage Capacity, Gt
Sligo and Hosston Formations and Cotton Valley Group	67.8	91.5	745.0	296	37.25
Sligo and Hosston Formations and Cotton Valley Group Deep	100	100	37.3	129	1.90
Yegua and Cockfield Formations	95	98	47.0	114	2.35
Frio and Vicksburg Formations	43	66	74	66	3.70
Lower Miocene I	92.5	99.5	39.6	50	1.98
Lower Miocene II	43	76	31.0	28	1.55
Middle Miocene	100	100	13.67	24	0.68
Upper Miocene	97	99.7	22.7	12	1.13
Carrizo Sand and Wilcox Group	100	100	96.64	246	4.83
Total			1,107.0	965	55.50

A_a: Percentage of area with CO_2 injection density of at least 0.25 Mt/year; I_a: percentage of CO_2 injection rate that takes place in A_a; V_t: total CO_2 injection volume per SAU per year, Mt/year; N_w: number of wells required to inject CO_2 in each SAU.

Sligo and Hosston Formations and Cotton Valley Group SAU in the U.S. Gulf of Mexico Basin. Note that hydrocarbon-bearing reservoirs and formations are divided into conventional and unconventional reservoirs, and that only conventional reservoirs have been investigated here—we did not identify or account for any unconventional hydrocarbon accumulations within each SAU. For example, based on the EIA (Energy Information Administration) (2010) evaluation, at least four tight-gas "plays" are accounted for within the Travis Peak (correlative to the Hosston Formation in Louisiana and Mississippi) and Cotton Valley formations. However, it was not possible for us to identify what portions of these plays are within the Sligo and Hosston Formation and Cotton Valley Group SAU.

Studies show that there could be several undiscovered hydrocarbon reservoirs within these SAUs in the U.S. Gulf of Mexico Basin (Schenk and Viger, 1995; Dyman and Condon, 2006; Warwick, 2017; Swanson and Karlsen, 2009; Dubiel et al., 2011; and Hackley, 2012) that further complicate the issue. Because hydrocarbon production from oil and gas reservoirs depends largely on each reservoir's prevailing conditions and operating pressure, any unplanned CO_2 injection into or near these reservoirs will disturb their governing and driving mechanisms that may have some undesired effects on the hydrocarbon reservoirs. Therefore, to account for the presence of oil and gas reservoirs, first all conventional oil and gas reservoirs within each SAU were extracted from the CRD (Carolus et al., 2017) and then the corresponding areal extents of these SAUs were plotted similar to Fig. 5 to evaluate what portion of each SAU is covered by the resident oil and gas reservoirs.

A series of decline curve analysis (DCA) of all of the oil and gas reservoirs within each SAU of the U.S. Gulf of Mexico Basin was performed based on the procedure explained elsewhere (Jahediesfanjani, 2017). Based on this analysis, the complete oil and gas production decline of the reservoirs within many of these SAUs may take a relatively long time (over 40 years on average) which further complicates the CO_2 injection and storage operations and considerations throughout these SAUs.

To account for the existence of hydrocarbon reservoirs in each SAU, the SAU is divided into 25 km by 25 km cells and the cells containing oil and gas reservoirs were excluded from the available CO_2 injection and storage volume calculation. Moreover, to create a buffer zone around the hydrocarbon-bearing cells, the CO_2 injection rates in the cells surrounding these reservoirs were lowered, or in some cases zeroed out, so

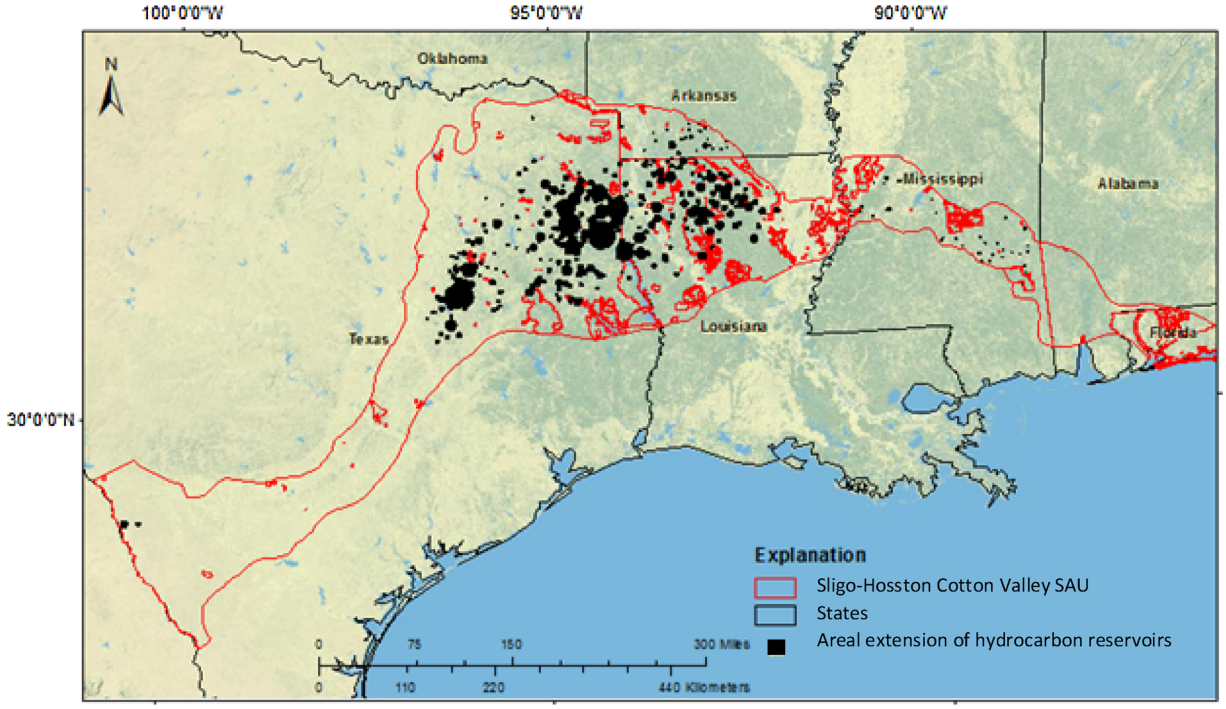


Fig. 5. The areal extent of the conventional oil and gas reservoirs within the Sligo and Hosston Formations and the Cotton Valley Group SAU (storage assessment unit) in the U.S. Gulf of Mexico Basin. The areal extents of the oil and gas reservoirs are from the Comprehensive Resource Database (CRD) (Carolus et al., 2017).

Table 4
The cumulative area and volume percentage of hydrocarbon reservoirs and their effect on the total CO_2 injection and storage volume of the SAUs (storage assessment units) within the U.S. Gulf of Mexico Basin. SAUs and most likely input calculation values are from U.S. Geological Survey Geologic Carbon Dioxide Storage Resources Assessment (USGS CO_2 assessment team, 2013a and 2013b).

#	SAU Name	Area, km^2	V, %	V_t, Mt/year	N_w	50-year CO_2 Storage Capacity, Gt
1	Norphlet Formation	1.29	0.02	20.14	188	1.007
2	Norphlet Formation Deep	102.5	0.3	6.26	262	0.31
3	Smackover Formation	1,836.2	0.92	1.65	55	0.08
4	Smackover Formation Deep	1,212	0.9	8.25	330	0.41
5	Haynesville Formation	862.1	0.53	0.8	46	0.04
6	Haynesville Formation Deep	86.4	0.34	0.70	50	0.03
7	Sligo and Hosston Formations and Cotton Valley Group	40,190	0.7	335	122	17.0
8	Sligo and Hosston Formations and Cotton Valley Group Deep	1,224	0.06	22.1	78	1.14
9	Knowles and Winn Limestones and Calvin Sandstone	N/A	N/A	N/A	N/A	N/A
10	Rodessa Formation and James Limestone	34,900	36.5	0.70	37	0.09
11	Rodessa Formation and James Limestone Deep	128.0	0.18	0.42	40	0.021
12	Fredericksburg Group and Rusk Formation	970	0.74	4.41	71	0.22
13	Edwards, Glen Rose, and James Limestones	443	4.22	0.31	19	0.016
14	Washita and Fredericksburg Groups, Rusk Formation, and James Limestone	3,700	6.1	0.91	134	0.04
15	Washita and Fredericksburg Groups, Rusk Formation, and James Limestone Deep	119	2.7	0.14	14	0.005
16	Tuscaloosa and Woodbine Formations	7,000	0.35	40.0	330	2.02
17	Navarro, Taylor, and Austin Groups	7,960	1.8	38.0	210	1.90
18	Carrizo Sand and Wilcox Group	10,200	0.71	9.6	25	0.5
19	Queen City Sand	1345	2.70	2.52	39	0.13
20	Sparta Sand	152.4	0.27	7.1	84	0.36
21	Yegua and Cockfield Formations	6,032.6	0.49	7.1	19	0.35
22	Frio and Vicksburg Formations	21,500	2.8	5.6	8	0.3
23	Lower Miocene I	698	45.0	1.6	20	0.80
24	Lower Miocene II	702	0.76	10.85	10	0.54
25	Middle Miocene	3760	2.8	0.07	2	0.035
26	Upper Miocene	844	0.45	1.15	1	0.06
27	Tertiary Slope and Basin Floor	N/A	N/A	N/A	N/A	0.00
Total, Group 1 to Group 6				526.30	2247	27.46
Total, Group 3 to Group 6				393.07	285	20.725

Area: cumulative area of oil and gas reservoirs within each SAU; V: cumulative volume of oil and gas reservoirs as a percentage of total SAU volume; V_t : total CO_2 injection volume in each SAU per year; N_w: number of wells required to inject CO_2 in each SAU; N/A: not applicable due to insufficient data.

Table 5
CO_2 storage capacity, number of wells required to inject CO_2, and the area considered for CO_2 injection and storage for all storage assessment units (SAUs) in six Groups (1–6) and three scenarios (1–3). Scenario 1: including all SAUs; scenario 2: all SAUs excluding a hydrocarbon reservoir buffer; scenario 3: Only those SAUs with fewer than half of their area covered by hydrocarbon reservoirs.

Group	Scenario 1			Scenario 2			Scenario 3		
	V_t, Mt/year	N_w	A, MM km^2	V_t, Mt/year	N_w	A, MM km^2	V_t, Mt/year	N_w	A, MM km^2
Group 1	202.2	8,009	5.01	149.3	6,617	4.14	109.3	6,544	4.09
Group 2	271.0	1,976	1.23	200.6	1,489	0.93	131.8	1,481	0.9
Group 3	153.0	496	0.22	76.0	271	0.17	26.9	253	0.17
Group 4	571.0	956	0.26	247.9	416	0.26	225.4	367	0.26
Group 5	47.0	37	0.43	22.0	18	0.01	20.6	3	0.01
Group 6	825.8	342	0.22	336.8	140	0.09	0.8	1	0.09
Total	2070.0	11,816	7.37	1032.6	8,951	5.60	514.8	8649	5.52
Groups 3-6	1596.8	1,832	1.13	682.7	845.0	0.53	273.7	624	0.53

V_t: total annual CO_2 injection volume in each group; N_w: number of wells required to inject CO_2 in each group; A: area available for CO_2 injection.

that the induced pressure plume remains only within each specific cell and does not affect the hydrocarbon-producing reservoirs' operation and performance.

Table 4 indicates the cumulative area and volume percentage of the hydrocarbon reservoirs and their effect on the total CO_2 injection and storage volume of all the SAUs within the U.S. Gulf of Mexico Basin. Based on current evaluation, the total annual CO_2 injection volume of these SAUs is 526.3 Mt/year using 2247 CO_2 injection wells; it results in a cumulative 50-year CO_2 injection volume of 27.5 Gt for all groups of C_{ID}. The corresponding value for SAUs in Group 3 and above is 393 Mt/year using 285 CO_2 injection wells, and total 50-year CO_2 injection capacity of 20.7 Gt—69%, 74%, and 69% reductions, respectively, compared to the case where the entire pore volume of the SAU is available for CO_2 injection and storage (reported in Table 2). Note that the mentioned volume estimation method is performed solely for the analysis of available pore volume for the purpose of CO_2 injection and storage operations. Unless the regional oil and gas producers and the CO_2 injection and storage operators reach a comprehensive CO_2 injection plan and agreement, such CO_2 injection in practice might be very complex and ultimately non-operational.

3.2. Pressure-limited (dynamic) CO_2 injection and storage capacity of all SAUs in the United States

The goal of the current section is to summarize the pressure-limited dynamic CO_2 storage capacity in all 192 SAUs under three scenarios: 1) regardless of the presence of hydrocarbon reservoirs, 2) excluding the volume of hydrocarbon reservoirs and their surrounding no-injection buffer zones, and 3) excluding SAUs that contain large numbers of hydrocarbon reservoirs (defined as SAUs with more than half of their area covered by hydrocarbon reservoirs).

A similar procedure to that explained above for the U.S. Gulf of

Mexico Basin was applied to the remaining 165 SAUs within 32 basins; the results are summarized and presented in Appendix 2. Based on these results, Table 5 lists the 50-year CO_2 injection rate (V_t), number of required wells to be drilled, and the corresponding total saline area available for CO_2 injection and storage in each of the six CO_2 injection groups and the scenarios of accounting for the presence of hydrocarbon reservoirs. According to Table 5, the total estimated V_t for all studied SAUs is 2070.0 Mt/year, 1032.6 Mt/year, and 514.8 Mt/year for all groups in scenarios 1–3, respectively. The number of wells required to inject the listed volumetric CO_2 rates is 11,816, 8,951, and 8649 wells for scenarios 1–3, respectively. The corresponding available area for CO_2 injection and storage is 7.37 million km^2, 5.60 million km^2, and 5.52 million km^2 for scenarios 1–3, respectively. However, note that the Groups 1 and 2 CO_2 injection, corresponding to C_{ID} of less than 250,000 metric tonnes/year, might not be economically feasible at the industrial scale. Therefore, the total estimated V_t for Groups 3 and above is 1596.8 Mt/year, 682.7 Mt/year, and 273.7 Mt/year for scenarios 1–3, respectively.

4. Discussion and analysis of results

Due to the complexity of SAUs containing large numbers of hydrocarbon reservoirs (SAUs with more than half of their area covered by hydrocarbon reservoirs), and because dividing these SAUs into smaller sections requires detailed geological analyses that are currently impossible due to insufficient data, the best approach may be to exclude them from CO_2 storage consideration until the complete decline of their oil and gas production. However, if ignoring these SAUs is impossible due to political or economic circumstances, our preliminary analysis (Table 5, scenario 2) indicates that the non-hydrocarbon sections of these SAUs, in conjunction with other non-hydrocarbon-bearing SAUs, may be able to handle around 1.0 Gt/year of CO_2 injection for the next 50 years, which will be sufficient to accommodate the mandated CO_2 volume based on the B2DS requirements in the United States (IEA, 2017) without needing to use a pressure management technique such as brine extraction. Such large CO_2 storage volume utilization will require detailed geological and engineering analysis of these SAUs and meeting several technical and non-technical requirements, such as the cooperation and agreement of stakeholders, including oil and gas operators.

Because meeting all these requirements may be very difficult, and due to the technical issues such as difficulties in assigning correct CO_2 injection well locations and rates especially in the vicinity of hydrocarbon reservoirs due to lack of sufficient detailed engineering and geological knowledge of a given SAU, the CO_2 storage operation in the SAUs containing large numbers of oil and gas reservoirs may not be practically achievable. Therefore, we should mainly focus on the SAUs without significant numbers of hydrocarbon-bearing formations (SAUs with less than half of their area covered with hydrocarbon reservoirs) and that have relatively high CO_2 injection and storage potential (Groups 3 and higher). The 16 SAUs in the United States that meet these criteria are listed in Table 6. Seven of these SAUs show 50-year annual injection rates (V_t) of less than 5.0 Mt/year, which indicates that those SAUs should also be excluded from the available saline pore volume considered for CO_2 storage due to their very low CO_2 injection capacity per SAU. Moreover, the Lower Torok Formation SAU, in northern Alaska, may also be a poor candidate for CO_2 storage operation due to the high cost of CO_2 transportation. Excluding these eight SAUs, there are only eight remaining SAUs that are the best candidates for CO_2 injection and storage with a cumulative 50-year annual CO_2 injection rate (V_t) of around 300 Mt/year, 1040 wells required, and available area of 650,000 km^2 (the first eight SAUs in Table 6 that are indicated in bold).

Obviously, the estimated volume is insufficient to accommodate the volume of CO_2 storage required under the B2DS scenario. However, the estimated volume is the ultimate volume that can be stored in the saline formations in the United States without implementing any pressure management techniques such as brine extraction. The estimated storage volumes are based on the most likely properties from the USGS assessment (USGS CO_2 assessment team, 2013a and 2013b) and hence are considered the most likely CO_2 injection and storage capacities of these formations.

5. Summary and conclusions

In order to fully implement the B2DS requirements proposed by the 2015 Paris Climate Agreement, it is estimated that over 1.0 Gt of CO_2 needs to be captured and injected for sequestration purposes annually in the United States (IEA, 2017). Assuming that all other requirements, including CO_2 capture and transportation, public and stakeholder awareness, and all economic and technical features, are in place to implement such large CO_2 injection volumes nationally, the main question becomes whether there is adequate underground saline capacity to accommodate the 1.0 Gt/year of captured CO_2 under various scenarios and conditions. Moreover, if implemented, how many CO_2 injection wells, and at what spacing, need to be drilled to be able to inject the estimated volumes? To answer these questions, we estimated the CO_2 injection and storage capacity for the 192 SAUs defined by the USGS national CO_2 storage assessment (USGS CO_2 assessment team, 2013a and 2013b). We did this using our developed pre-brine extraction model (Jahediesfanjani et al., 2017) that is based on applying 3D numerical simulation only on the effective injection area surrounding each injection well that can practically accommodate the injected CO_2 volume while keeping the maximum reservoir pressure below 80% of the formation fracture pressure.

Because many of the saline formations in these SAUs have more than half of their area covered by actively produced conventional or unconventional hydrocarbon reservoirs, any large-scale CO_2 injection in these saline formations might severely affect these hydrocarbon reservoirs and their current and future performance. To alleviate the problem, three scenarios of saline formation CO_2 storage capacity estimates were calculated: 1) ignoring the presence of hydrocarbon reservoirs, 2) assuming no CO_2 injection in the areas containing hydrocarbon reservoirs and the surrounding buffer zone, and 3) injecting CO_2 only in those SAUs with less than half of their area covered by actively produced conventional or unconventional hydrocarbon reservoirs. The estimated CO_2 storage capacity volumes under each scenario are 1597 Mt/year, 683 Mt/year, and 274 Mt/year, respectively, for SAUs that show most likely CO_2 injection rates of at least 250,000 metric tonnes per year per well (Group 3 and higher).

The results indicate that the CO_2 storage capacity estimation based on scenario 2 may be able to accommodate the required CO_2 injection and storage according to B2DS. However, such capacity estimation is theoretical, and in practice many factors must come together, including detailed geological, geophysical, and engineering analyses of the SAUs containing hydrocarbon reservoirs to ensure that neither CO_2 plumes nor induced pressure fronts interrupt oil and gas production. Also needed are comprehensive plans to perform a CO_2 injection project and an agreement among all players, including the present oil and gas operators. As having all these requirements come together may be very difficult, the only other feasible option is CO_2 storage based on scenario 3. Excluding smaller SAUs and Alaskan SAUs, there are only eight major SAUs in scenario 3 with considerable CO_2 storage capacity and an insignificant number of hydrocarbon reservoirs; these have a total CO_2 injection and storage capacity of less than 300 Mt/year. As the estimated CO_2 storage capacity based on scenario 3 seems insufficient to handle B2DS and even 2DS scenarios, these selected SAUs can be considered as a starting point for CO_2 storage operations. Further CO_2 injection is certainly possible by applying pressure management techniques, such as brine extraction, that will be considered in our future studies.

Table 6
Storage assessment units (SAUs) with fewer than half of their area covered by hydrocarbon reservoirs: annual CO_2 injection capacity, available area, and number of wells required (groups 2–7). SAUs are from U.S. Geological Survey Geologic Carbon Dioxide Storage Resources Assessment (USGS CO_2 assessment team, 2013a and 2013b). First eight SAUs that are in bold are selected for further analysis.

#	SAU	Basin	Average C_{ID}, Mt/year /Well	Acceptable Injection, %	V_t, Mt/year	N_w	Area, km^2
1	**Mount Simon Sandstone**	**Illinois Basin**	**0.16**	**80**	**29.44**	**230**	**143,393.30**
2	**Ordovician Composite**	**Illinois Basin**	**0.7**	**100**	**43.4**	**62**	**38,164.1**
3	**Basin Flank Paleozoic Composite**	**Palo Duro Basin**	**0.27**	**100**	**5.4**	**20**	**12,375.73**
4	**Pre-Punta Gorda**	**South Florida Basin**	**0.69**	**100**	**93.15**	**135**	**83,809.32**
5	**Cedar Keys and Lawson Formations**	**South Florida Basin**	**0.14**	**91**	**18.60**	**146**	**90,871.34**
6	**Norphlet Formation**	**U.S. Gulf Coast**	**0.11**	**75**	**15.43**	**187**	**116,766.90**
7	**Navarro, Taylor, and Austin Groups**	**U.S. Gulf Coast**	**0.16**	**65**	**25.38**	**244**	**152,381**
8	**Eocene Composite**	**Western Oregon and Washington Basins**	**2.25**	**100**	**31.5**	**14**	**8,620.11**
9	Lower Torok Formation	Alaska North Slope	0.52	100	77.48	149	92,886.74
10	Upper Torok Formation	Alaska North Slope	0.15	90	3.24	24	14,973.90
11	Staines Tongue	Alaska North Slope	0.36	100	4.68	13	7,956.40
12	Hilliard, Baxter, and Mancos Shales	Greater Green River Basin	0.36	100	4.68	13	30,850.28
13	Mesaverde Group	Greater Green River Basin	0.25	38	4.75	50	16,098.97
14	Paleozoic Composite	Wyoming-Idaho-Utah Thrust Belt	0.26	70	4.73	26	11,983.17
15	Nugget Sandstone	Wyoming-Idaho-Utah Thrust Belt	0.27	90	4.86	20	15,115.55
16	Nugget Sandstone Deep	Wyoming-Idaho-Utah Thrust Belt	0.25	75	4.69	25	14,591.05
Total					371.4	1,369	850,837.9
Total remaining 8 SAUs excluding Alaska and $V_t < 5.0$ Mt/year					262.3	1,037	646,400.0

C_{ID}: CO_2 injection density; Acceptable Injection: percentage of the SAU injection capacity that fits within group 3 and above; V_t: total annual CO_2 injection volume in each SAU; N_w: number of wells required to inject CO_2 in each SAU.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijggc.2018.09.011.

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